

Motion Terms

- Distance
- Displacement
- Speed
- Velocity
- Rate
- Acceleration
- momentum

Motion?

- **Motion** is the change in position of an object with respect to time.
- **Motion** is typically described in terms of velocity, acceleration, displacement and time.

Displacement v Distance

Distance

- is the total length of the path of motion
- Scalar quantity- has size and no direction.

Displacement

- is the linear distance between the initial and final point of an object
- Vector quantity- has both size and direction 

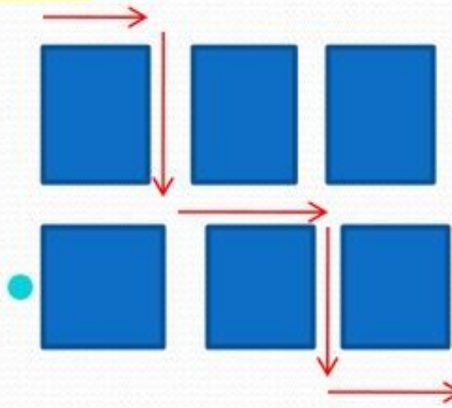
Vector or Scalar?

- 5 m
- 30 m/sec, East
- 5 m, North
- 20 degrees Celsius
- e. 256 bytes
- f. 4000 Calories

Calculations: Example 1



home



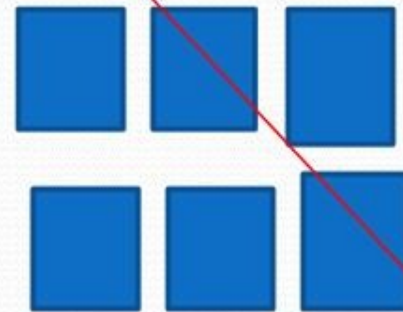
school



$$\begin{aligned}\text{Distance} &= 1.2 + 2 + 2 + 2 + 1.2 \\ &= 8.4\text{m}\end{aligned}$$



(starting point)



(end point)



$$\text{Displacement} = 7.4\text{m}$$

south east

Speed vs. Velocity

Speed

- Measure of how quickly something moves
- Scalar quantity
- Speed can be measured in different units. E.g . m/s, km/h, km/s, miles per hour.

Conversion:

- speed in km/h $\xrightarrow{\div 3.6}$ speed in m/s
- Speed in m/s $\xrightarrow{\times 3.6}$ speed in km/h

Velocity

- The rate at which an object changes its position.
- Vector quantity
- The direction of the velocity is simply the same as the direction that an object is moving.
- E.g airplane moving towards the west with a speed of 300 mi/hr has a velocity of 300 mi/hr, west
- Average velocity =
 $\text{Position/time} = \text{displacement/time}$



Question 1

- Convert the following into the standard units (metres and seconds):
(a) 3 km (b) 37 cm (C) 3mins

Solution

- a) 3000m
- b) 0.37m
- c) 180seconds

QUESTION 2

- Convert the following times into the units in brackets:

(a) 300 s (min)

(b) 9 hours (min)

(c) 750 min (hours)

Solution

a) 5mins

b) 540mins

c) 12.5hrs

Question 3

- A small car can top speed at 180 km/h. Write this in SI units (m/s).
- Convert 3m/s into **km/h**

Solution

- Convert **km/h** into **m/s** by $\div 3.6$

Therefore: $180/3.6 = 50\text{m/s}$

- Convert **m/s** into **km/h** by $\times 3.6 =$
- $3 \times 3.6 \text{ km/h} = 10.8 \text{ km/h}$

QUESTION 4

- A taxi drives 360km in 4 hours.

(a) What is the average speed?

(b) How long will it take to drive 540km at the same speed?

Solution

- Average Speed = $\frac{\text{Total Distance (km)}}{\text{Total Time (hr)}}$
 $= 360/4$
 $= 90\text{km/hr}$

$$\begin{aligned}\text{Time taken} &= \frac{\text{Distance}}{\text{Speed}} \\ &= 540/90 \\ &= 6\text{hrs}\end{aligned}$$

Question 5

Trinh rides her bike with a constant speed of 5 m/s . It takes her 3 minutes to get to the milk bar. Calculate how far away it is.

Solution

First, convert the time she took into seconds in order to state the answer in metres.

$$t = 3 \times 60 = 180 \text{ s}$$

Trinh has travelled:

$$d = v \times t$$

$$= 5 \times 180$$

$$= 900 \text{ m}$$

- The milk bar is 900 m away.

Question 6

Theo spent 8 hours travelling 400 km from his home in Bundaberg to visit his sister in Toowoomba. Calculate Theo's average speed for the journey.

Solution

$$\begin{aligned}\text{Speed (km/h)} &= \frac{\text{distance (km)}}{\text{time (hr)}} \\ &= 400/8 \\ &= 50\text{km/h}\end{aligned}$$

Calculating Speed & distance

$$\text{average speed} = \frac{\text{total distance travelled (m)}}{\text{total time taken (s)}}$$

or

$$v = d/t \text{ (m/s)}$$



Instantaneous Speed

- Speed at a particular instant.
- Why do you think instantaneous speed is important??

Velocity

- The rate at which displacement changes.
- Vector quantity
- Simply a speed with direction
- Average velocity=

$$\frac{\text{Change in Position}}{\text{Time}} = \frac{\text{Displacement}}{\text{Time}}$$

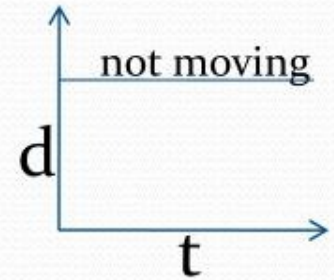
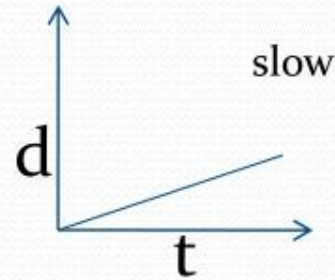
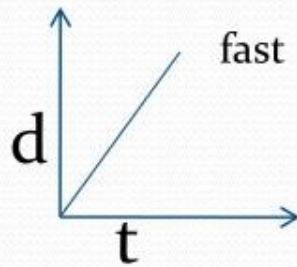
Graphing motion

- Distance-time graph
- Displacement-time graph
- Speed-time graph

} time is always placed
on the horizontal axis.

Distance-Time Graph-

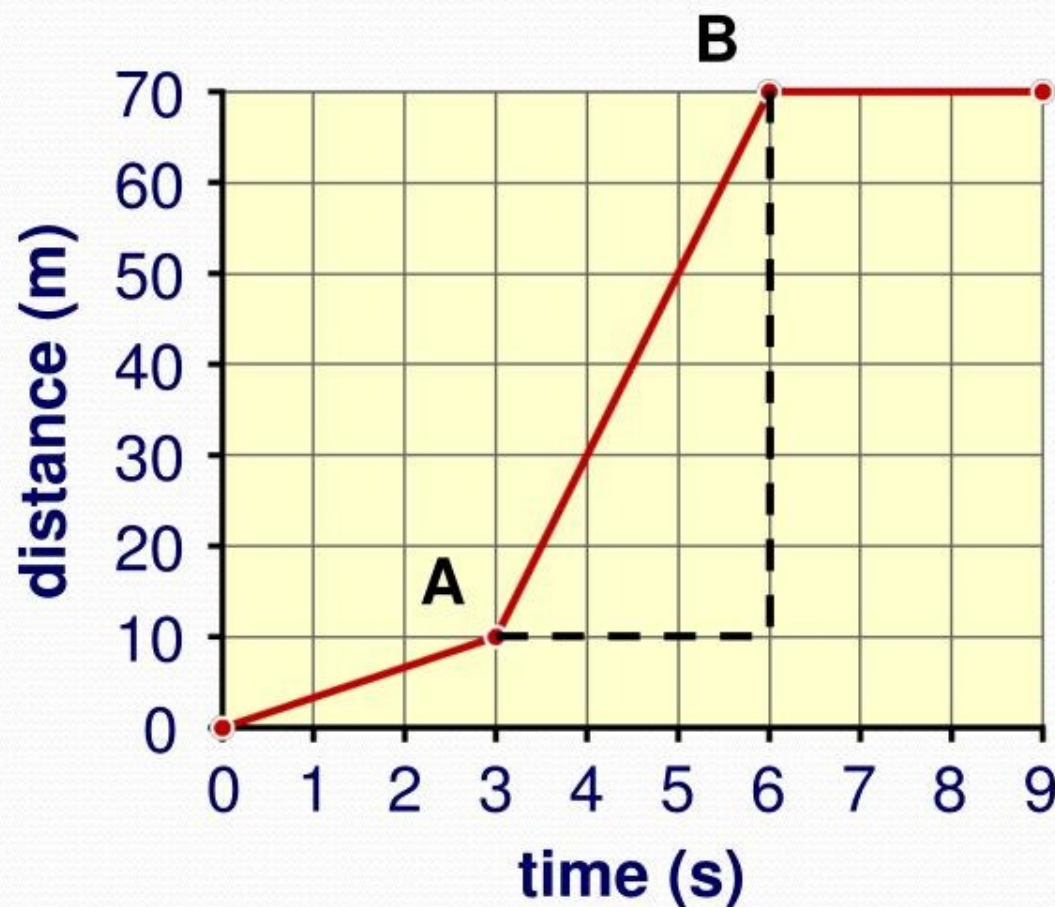
shows how far an object travels as time progresses.



- The steeper the gradient, the faster the object is moving.
- The slope or gradient of a distance-time graph is equivalent to **the object's average speed over a time interval** —————> choose two points to calculate the gradient and use the formula **RISE/RUN**

Example:

What is the speed of the object between points **A** and **B**?



- Choose two points to calculate the gradient

- Gradient = rise/run**

- the object has moved 60m (70 – 10)

- it took 3s to move this distance (6 – 3)

- speed = distance/time

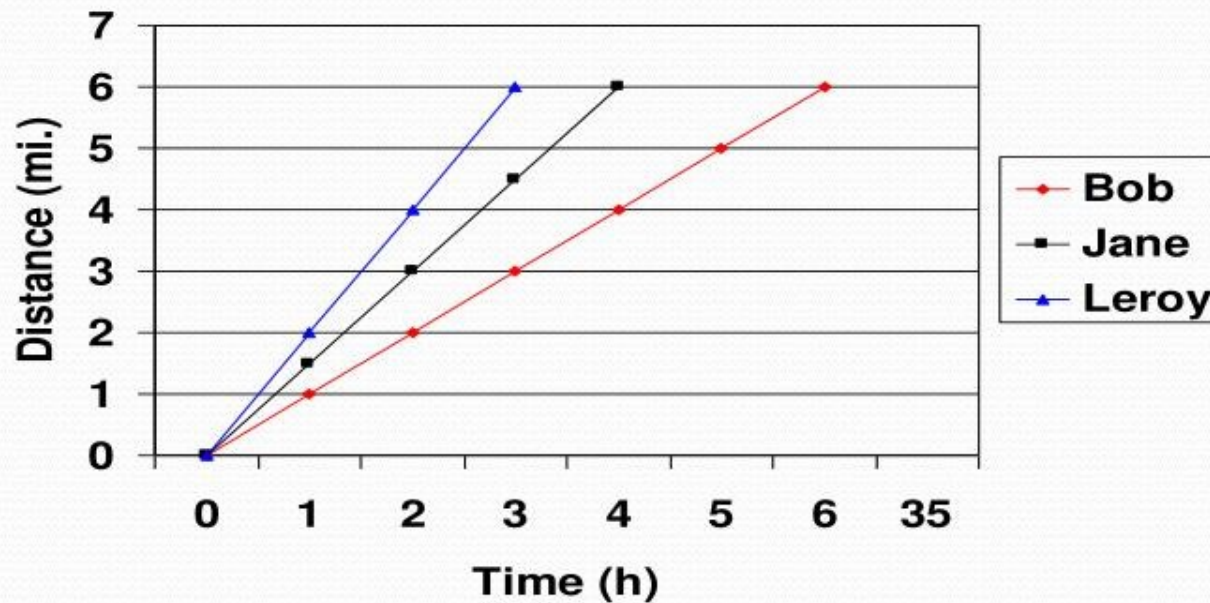
$$= 60/3$$

$$= \mathbf{20\,m/s}$$

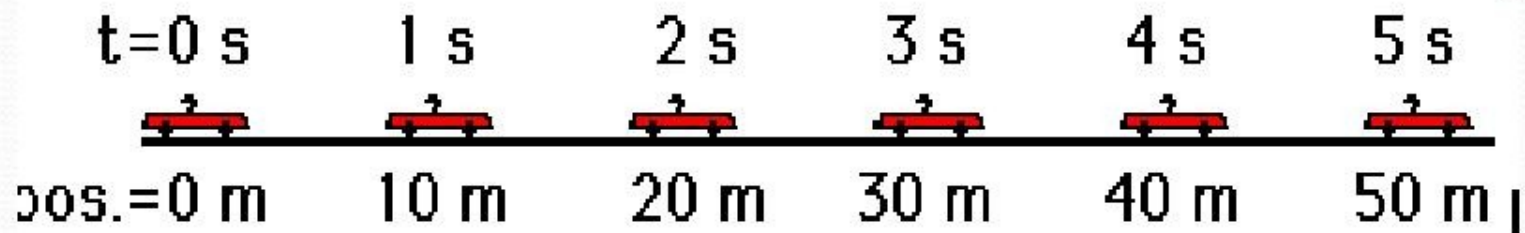


Question

- Below is a distance vs. time graph for 3 runners. Who is the fastest?

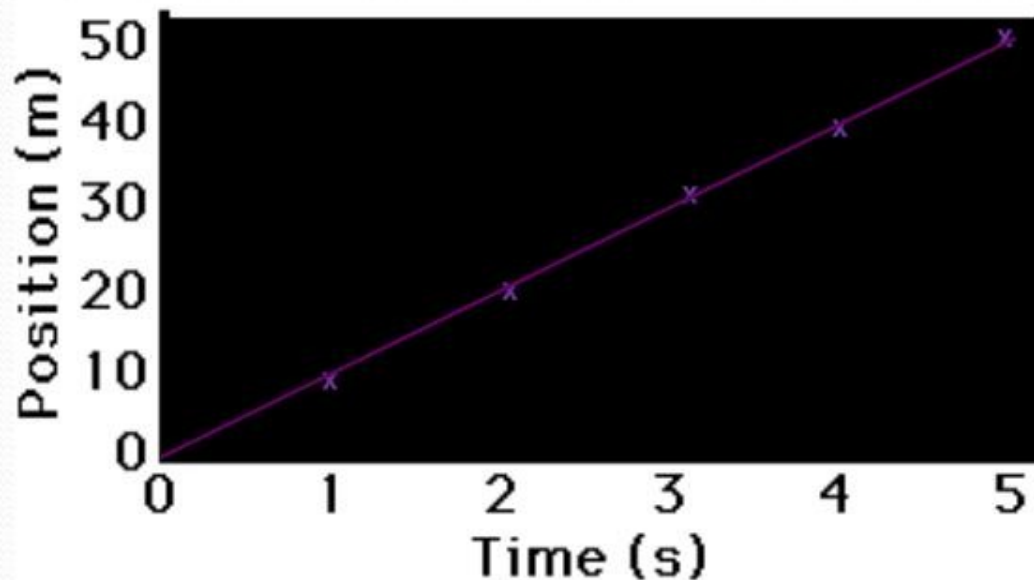


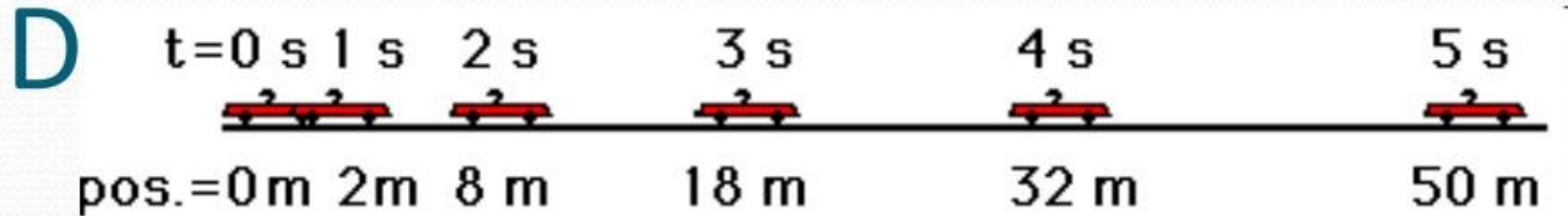
Distance v Time



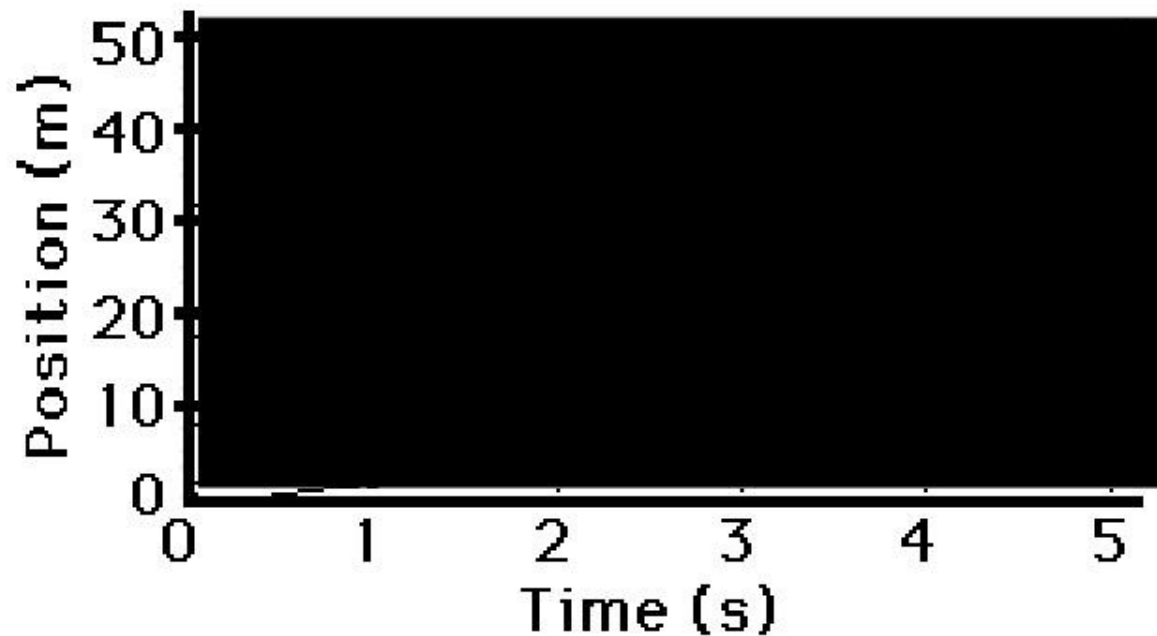
Graph the motion of the car.

Describe the motion?

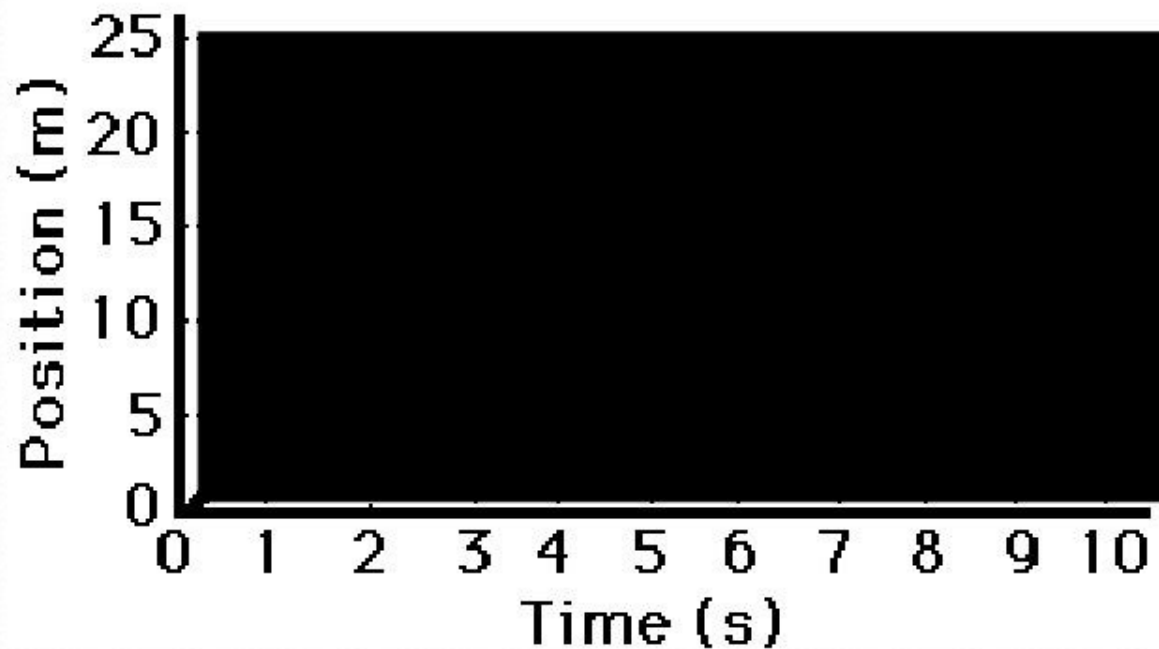
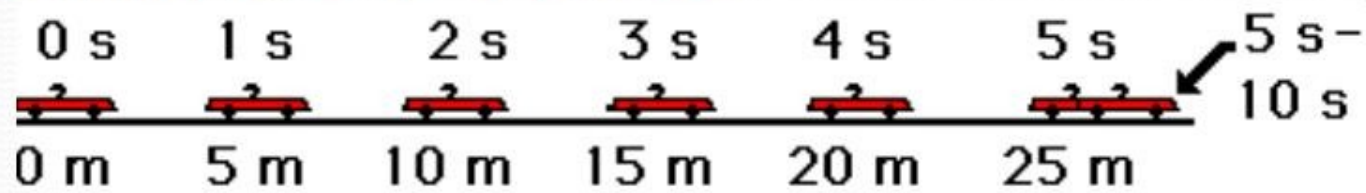




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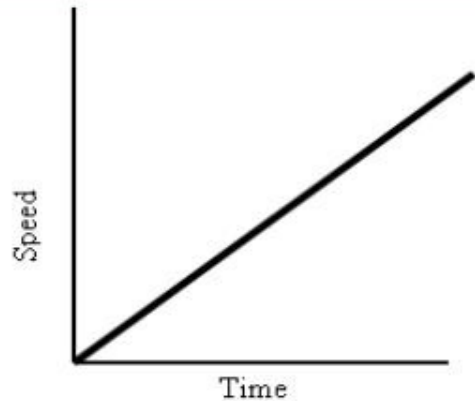


Distance v Time

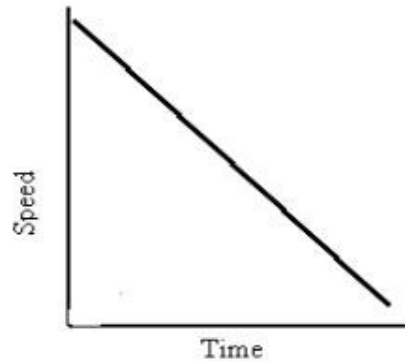


Speed-time graph

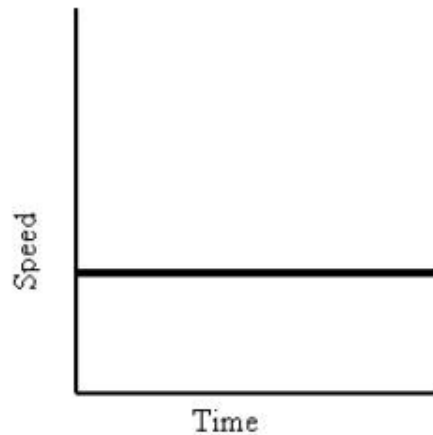
- Speed – time graph are also known as **velocity-time graph**
- A speed- time graph shows how an object's speed changes over time
- The area below a speed time graph is the distance the object has travelled up to a given point



This graph shows increasing speed.
The moving object is **accelerating**



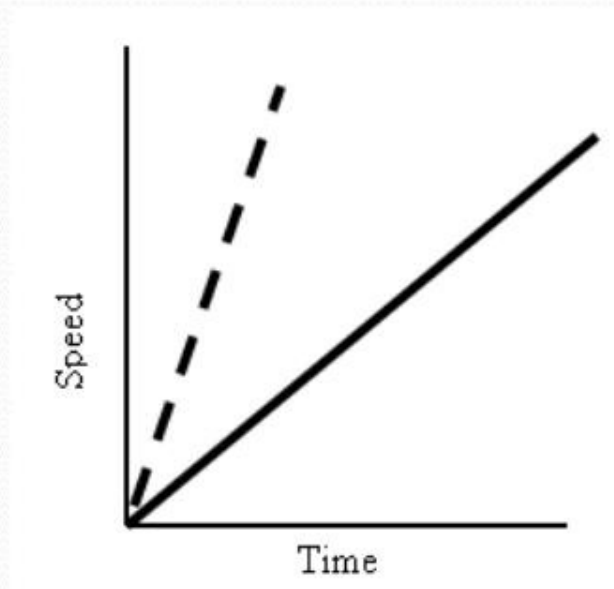
This graph shows decreasing speed.
The moving object is **decelerating**



A straight horizontal line on a speed-time graph means that speed is constant. It is not changing over time.

A straight line does not mean that the object is not moving

What about comparing two moving objects at the same time?

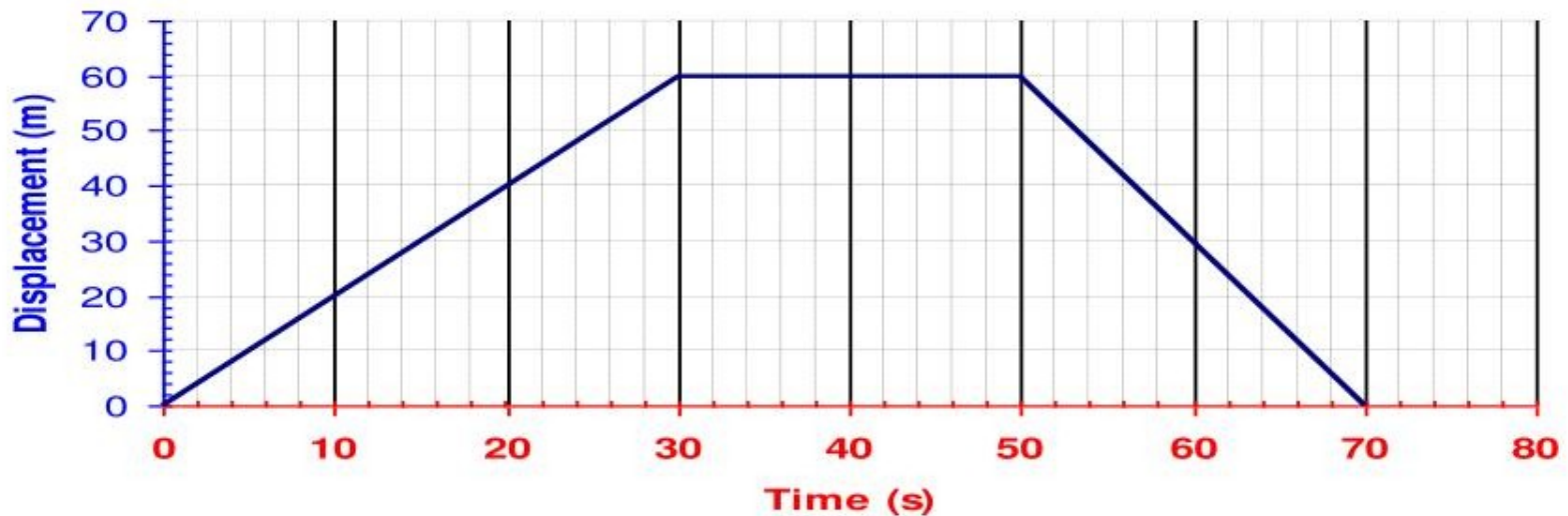


Answer:

- Both the dashed and solid line show increasing speed.
- Both lines reach the same top speed, but the solid one takes longer.
- **The dashed line shows a greater acceleration.**

Displacement – time graph

The displacement – time graph shows the journey of a woman going to a corner shop and back.



Calculate each of the following.

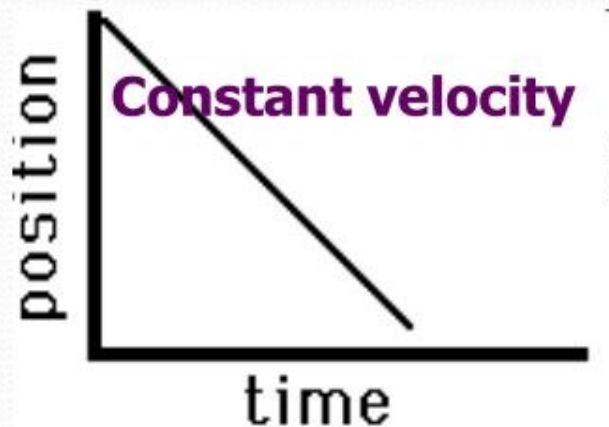
(a) Her total distance travelled.

(b) Her final displacement.

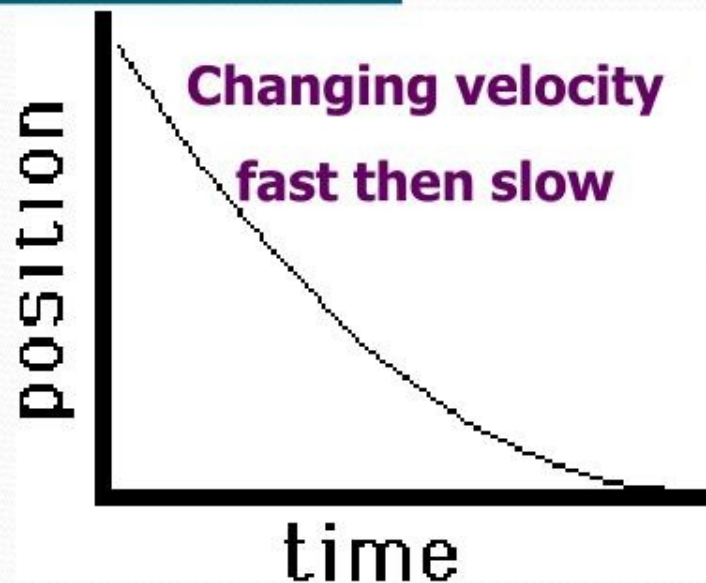
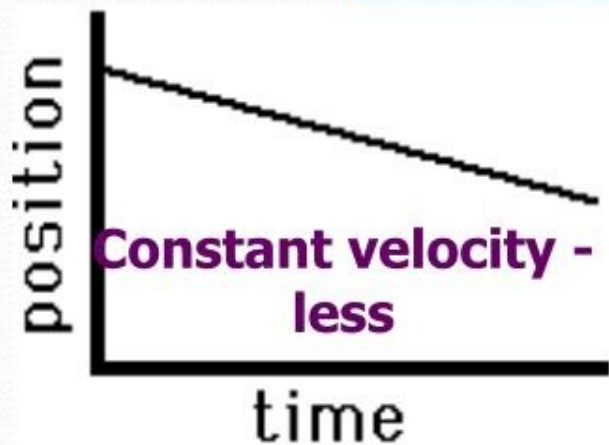
(a) $60 + 60 = 120\text{m}$

(b) 0

Distance v Time



Changing velocity is acceleration



Motion Formulas

$$V_{\text{ave}} = \frac{s}{t}$$

$$V_{\text{ave}} = \frac{(u + v)}{2}$$

$$a = \frac{(v - u)}{t}$$

$$s = ut + \frac{1}{2} a t^2$$

$$s = v.t - \frac{1}{2} a.t^2$$

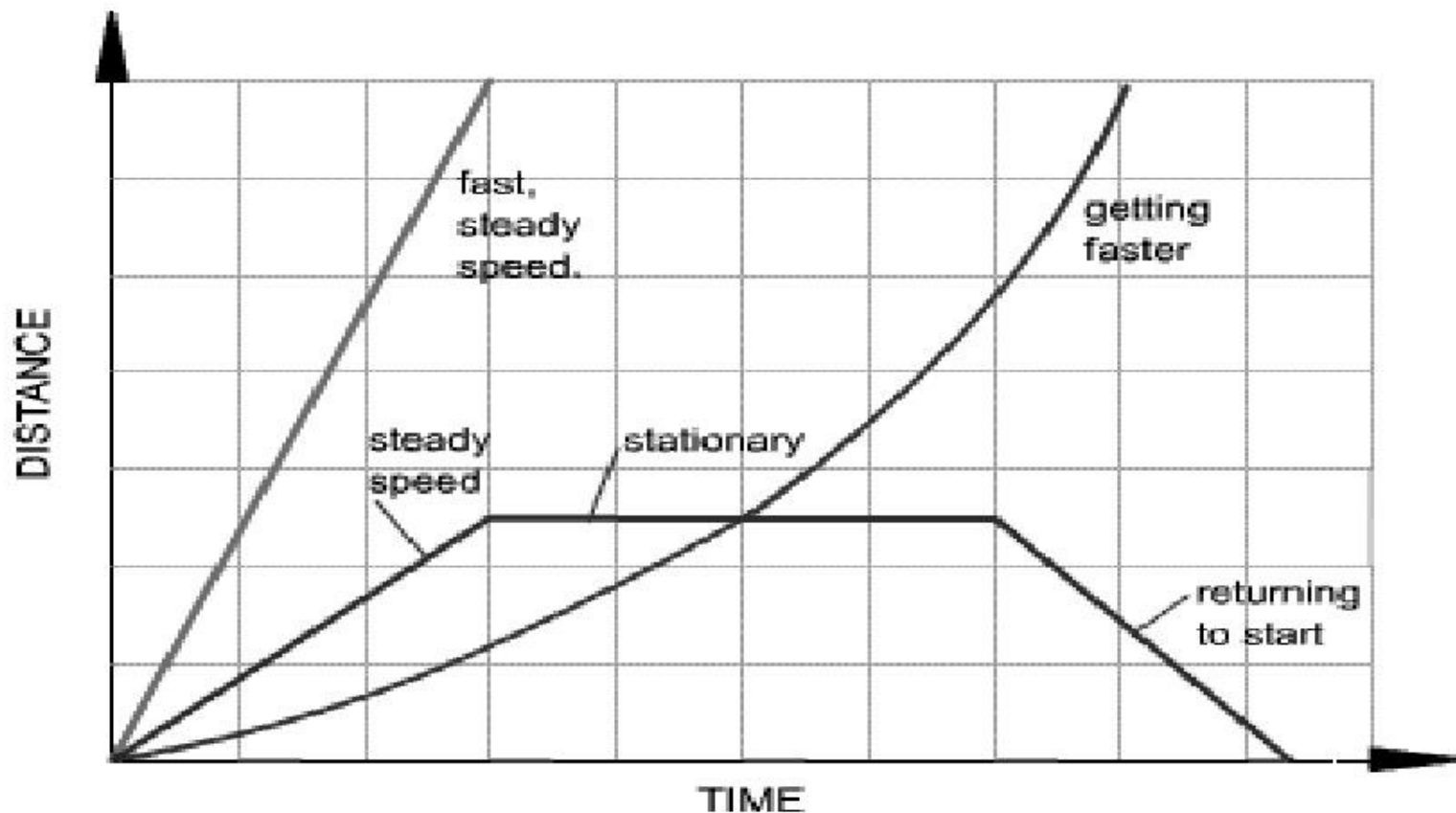
$$v^2 = u^2 + 2as$$

$$p = m . v$$

Summary:

A distance-time graph tells us how far an object has moved with time.

- The steeper the graph, the faster the motion.
- A horizontal line means the object is not changing its position - it is not moving, it is at rest.
- A downward sloping line means the object is returning to the start.

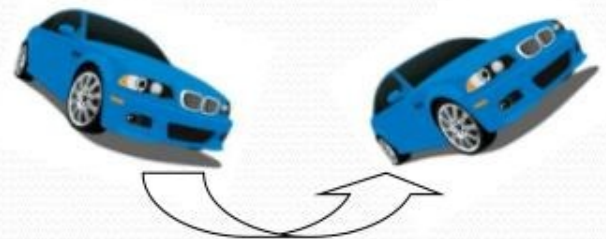


Acceleration

- **Acceleration** = speeding up
- **Acceleration** – the rate at which velocity changes
 - Can be an:
 - Increase in speed
 - Decrease in speed
 - Change in direction

Types of acceleration

- Increasing speed
 - Example: Car speeds up at green light
- Decreasing speed
 - Example: Car slows down at stop light
- Changing Direction
 - Example: Car takes turn (can be at **constant speed**)



Calculating acceleration



$$a = \frac{v - u}{t}$$

a = acceleration

v = final velocity

u = initial velocity

t = time taken

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

<http://www.one-school.net/Malaysia/UniversityandCollege/SPM/revisioncard/physics/forceandmotion/lineararmotion.html>

Acceleration

= change in velocity



Question

- How can a car be accelerating if its speed is a constant 65 km/h?
- If it is changing directions it is accelerating

Calculating Acceleration

- If an object is moving in a straight line

$$\text{Acceleration} = \frac{\text{Final_speed} - \text{Initial_Speed}}{\text{Time}}$$

Or

$$a = \frac{v-u}{t}$$

- Units of acceleration: m/s^2

Calculating Acceleration

$$\text{Acceleration} = \frac{\text{Final Speed} - \text{Initial Speed}}{\text{Time}}$$

$$= \frac{16 \text{ m/s} - 0 \text{ m/s}}{4 \text{ s}}$$

$$= 4 \text{ m/s}^2$$

0 s

1 s

2 s

3 s

4 s



0 m/s

4 m/s

8 m/s

12 m/s

16 m/s

Question

- A skydiver accelerates from 20 m/s to 40 m/s in 2 seconds. What is the skydiver's average acceleration?

$$\begin{aligned} \text{Accel} &= \frac{\text{Final_speed} - \text{Initial_speed}}{\text{Time}} \\ &= \frac{40\text{m/s} - 20\text{m/s}}{2\text{s}} = \frac{20\text{m/s}}{2\text{s}} \\ &= 10\text{m/s}^2 \end{aligned}$$

- The formula $a = \frac{v-u}{t}$ can be rearranged to allow the
- final speed of an object to be calculated:
- Final speed = initial speed + (acceleration x time)

Formula could be rearranged to find time

- $$\text{Time} = \frac{\text{Final speed} - \text{Initial Speed}}{\text{Acceleration}}$$

Problem 1:

- A roller coaster car rapidly picks up speed as it rolls down a slope. As it starts down the slope, its speed is 4 m/s. But 3 seconds later, at the bottom of the slope, its speed is 22 m/s. What is its average acceleration?

Solution

- Acceleration = $\frac{\text{final speed} - \text{initial speed}}{\text{time}}$

$$a = \frac{v - u}{t}$$

$$a = \frac{22-4}{3} = \frac{18}{3} = 6\text{m/s/s}$$

Problem:

- A train initially travelling at 30 km/h accelerates at a constant rate of 2 km/h/s for 30 seconds. Calculate its final speed.

Solution:

- Final speed = Initial speed + acceleration x time

$$v = u + at$$

$$v = 30 + (2 \times 30)$$

$$v = 30 + 60$$

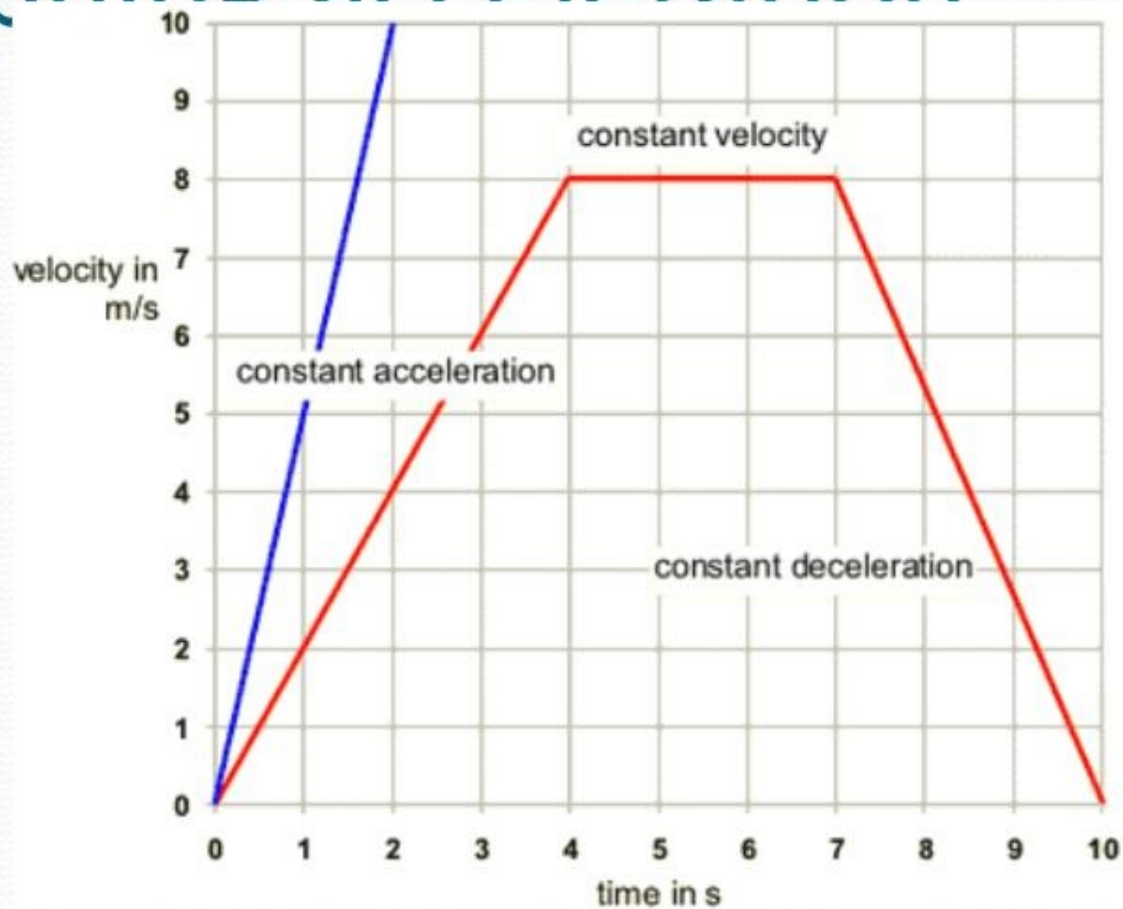
$$v = 90 \text{ km/h}$$

The train is travelling at 90km/h
after 30 seconds

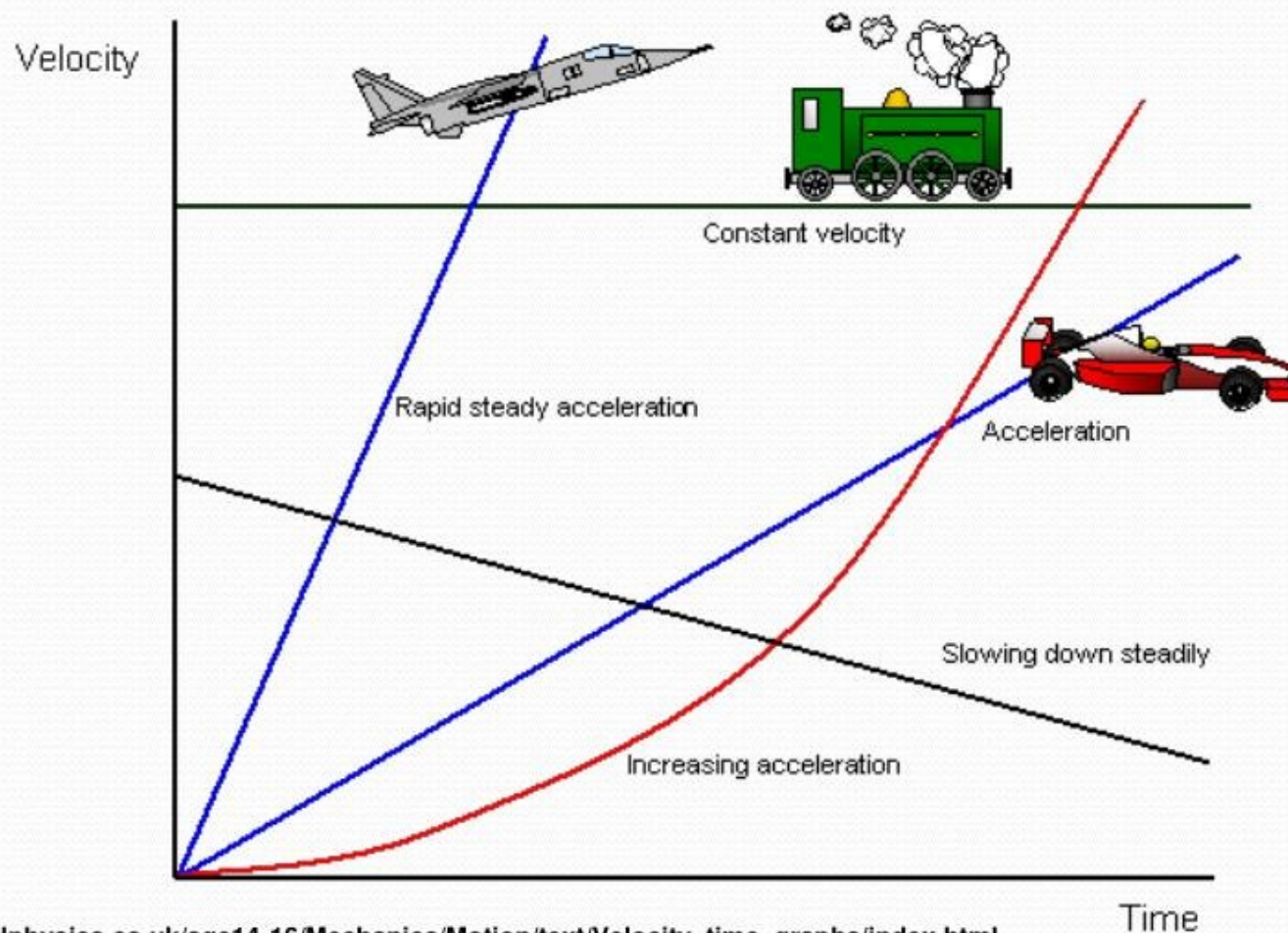
Graphing Acceleration

- Can use:
 - Velocity or speed – time graph= the acceleration can be calculated from the slope or gradient of a velocity/speed-time graph.

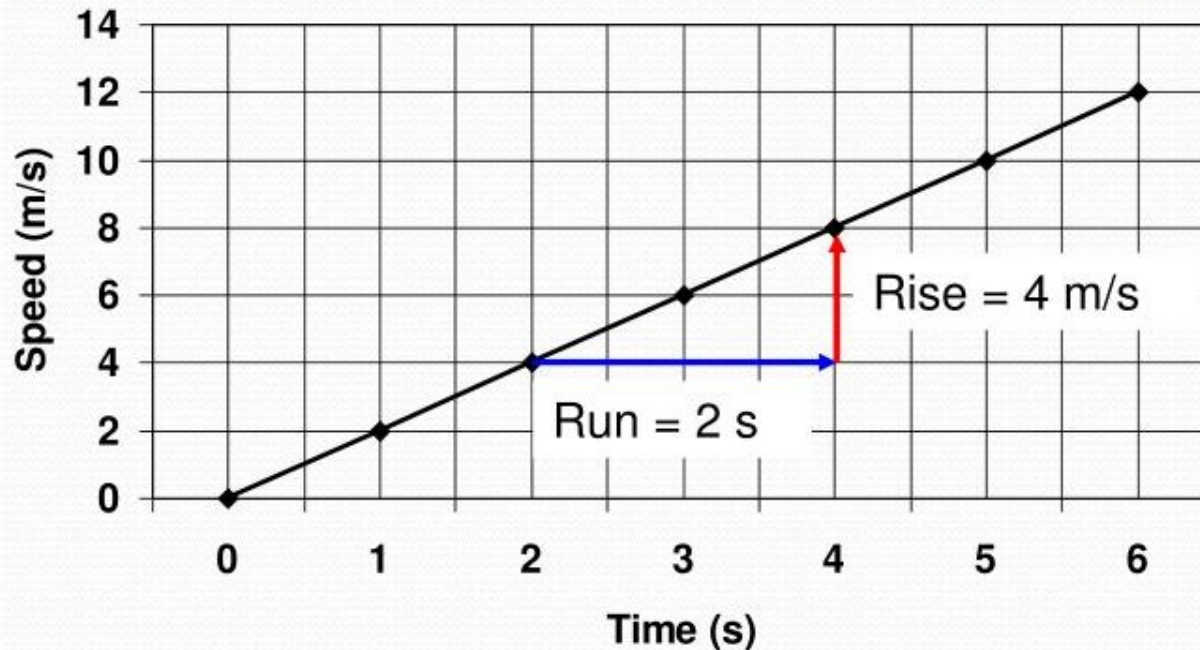
Graphing acceleration



The horizontal straight line shows something that is moving with a constant velocity. Straight lines slanting upwards show objects whose velocity is increasing at a steady rate – they have **constant positive acceleration**. Straight lines slanting downwards show objects whose velocity is decreasing at a steady rate – they have a **constant negative acceleration (retardation)**. The steeper the line the greater the acceleration or retardation. **A curved line shows an object whose acceleration is changing as time goes by.**



Graphing Acceleration: Speed vs. Time Graphs

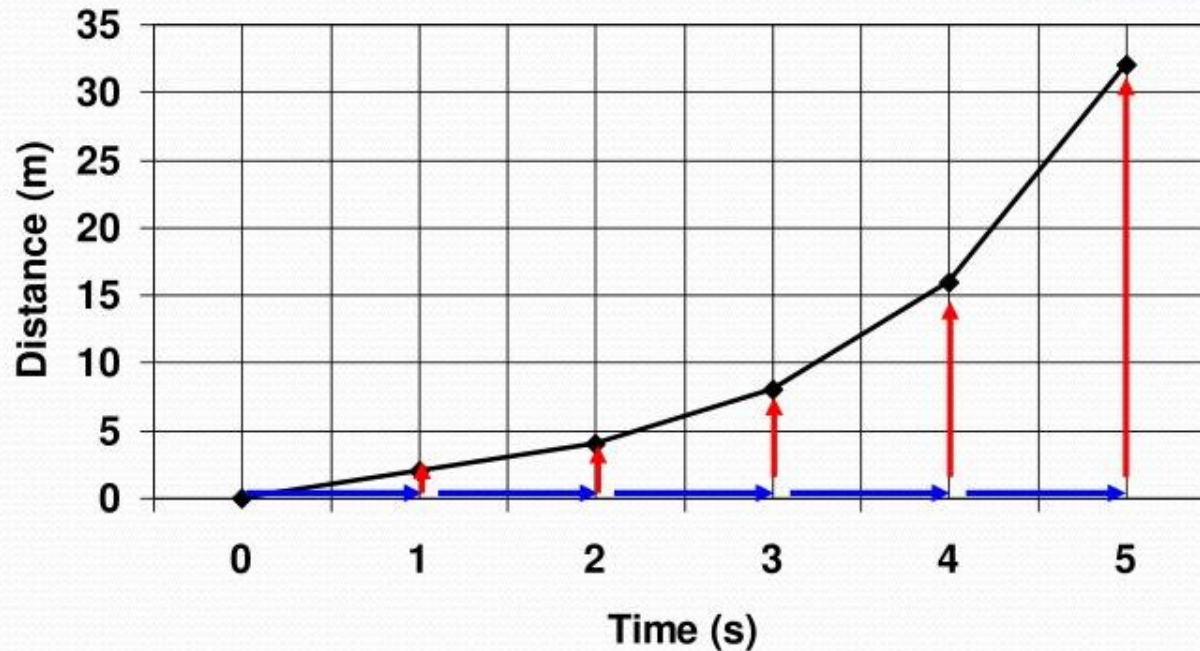


1) In Speed vs. Time graphs: How to calculate acceleration?

$$\begin{aligned}\text{Acceleration} &= \text{Rise/Run} \\ &= 4 \text{ m/s} \div 2 \text{ s} = \mathbf{2 \text{ m/s}^2}\end{aligned}$$

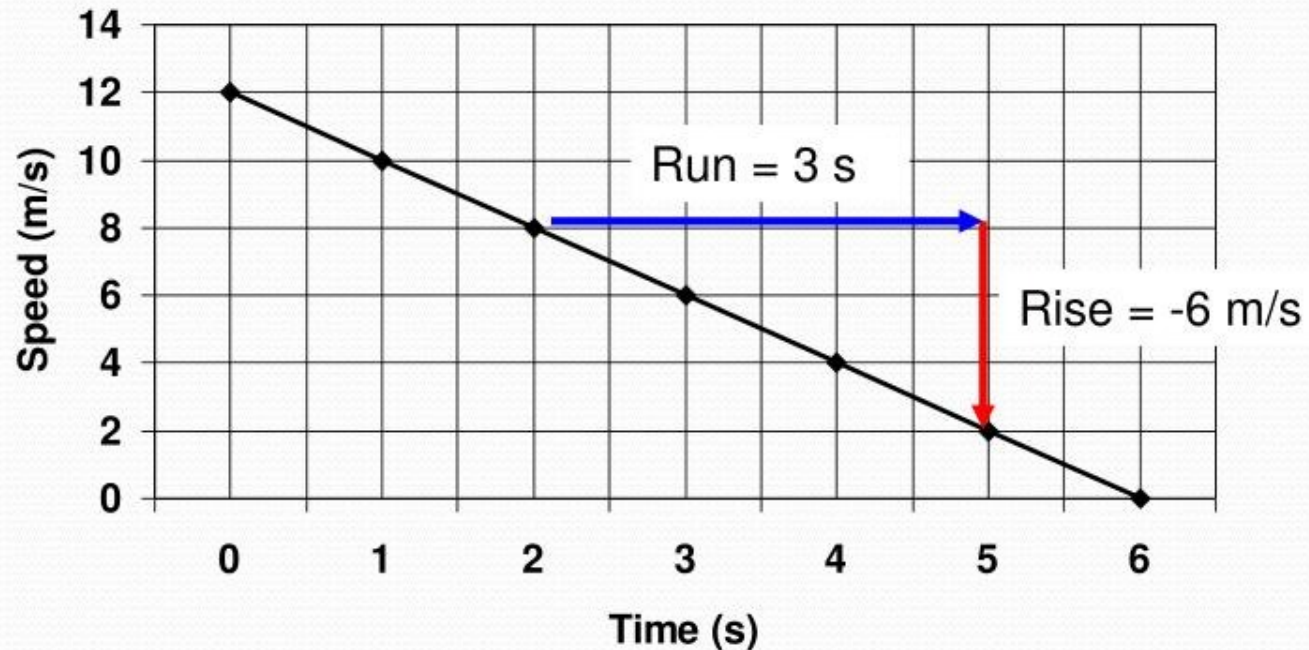
Graphing Acceleration:

Distance vs. Time Graphs



- 1) On Distance vs. Time graphs a curved line means the object is accelerating.
- 2) Curved line also means your speed is increasing. Remember slope = speed.

Question



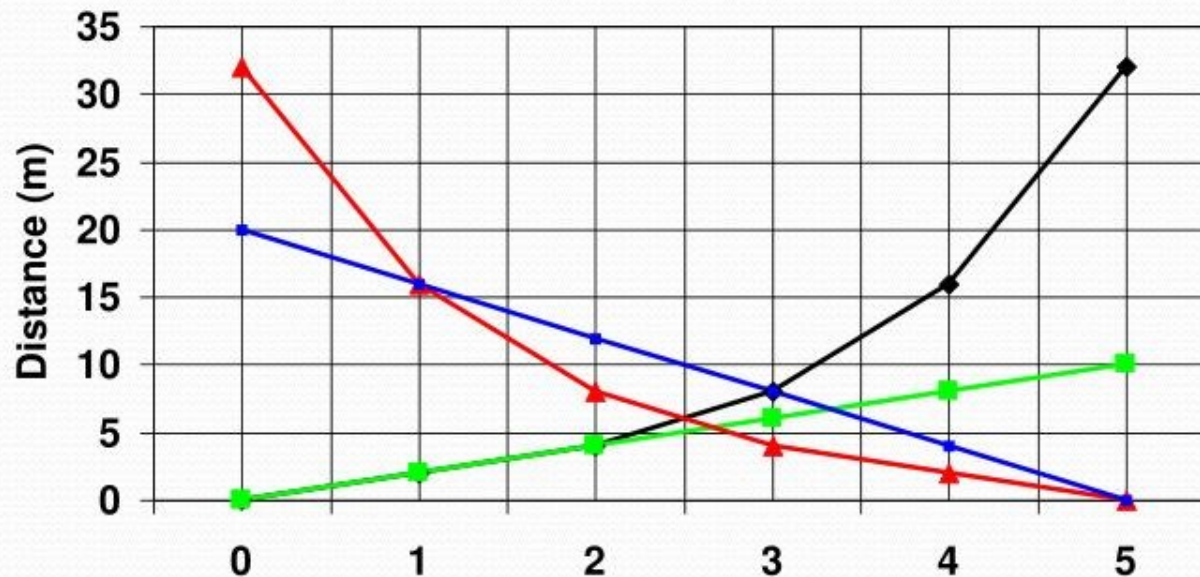
Above is a graph showing the speed of a car over time.

- 1) How is the speed of the car changing (speeding up, Slowing down, or staying the same)?
- 2) What is this car's acceleration?

Answers

- 1) The car is slowing down
- 2) Acceleration = rise/run = $-6\text{m/s} \div 3\text{s} = -2 \text{ m/s}^2$

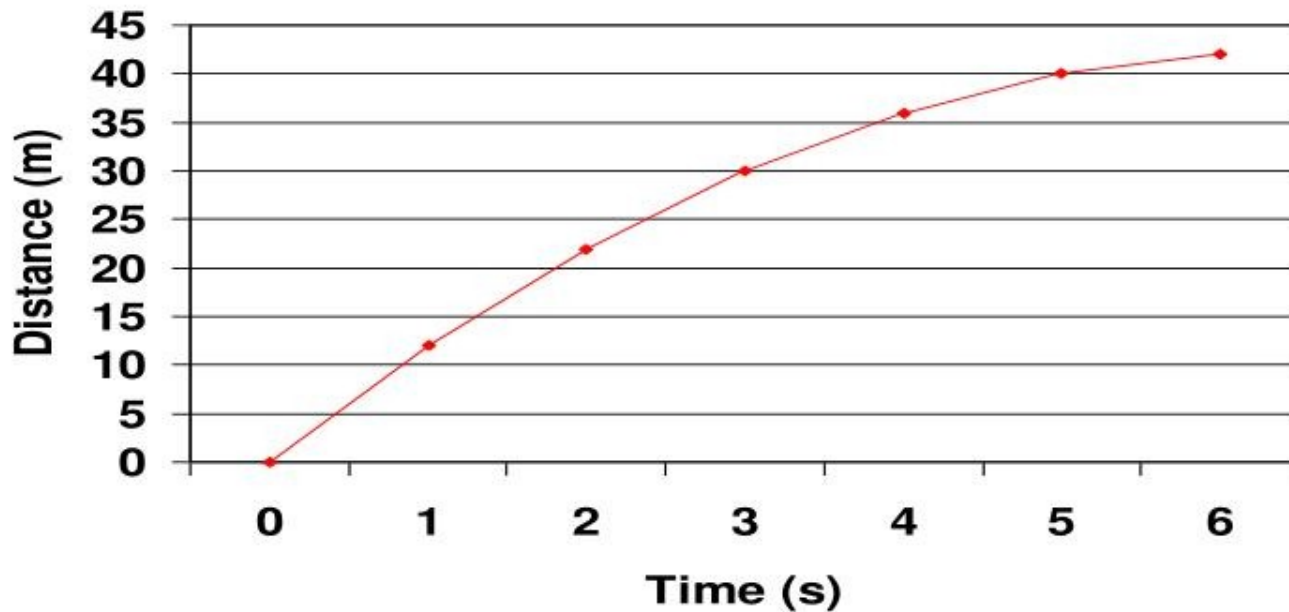
Question:



The **black line** represent a objects that are accelerating. Black is going a greater distance each second, so it must be speeding up. Red is going less each second, so must be slowing down

Remember: in distance vs. time graphs:
curved line = accelerating, flat line = constant speed

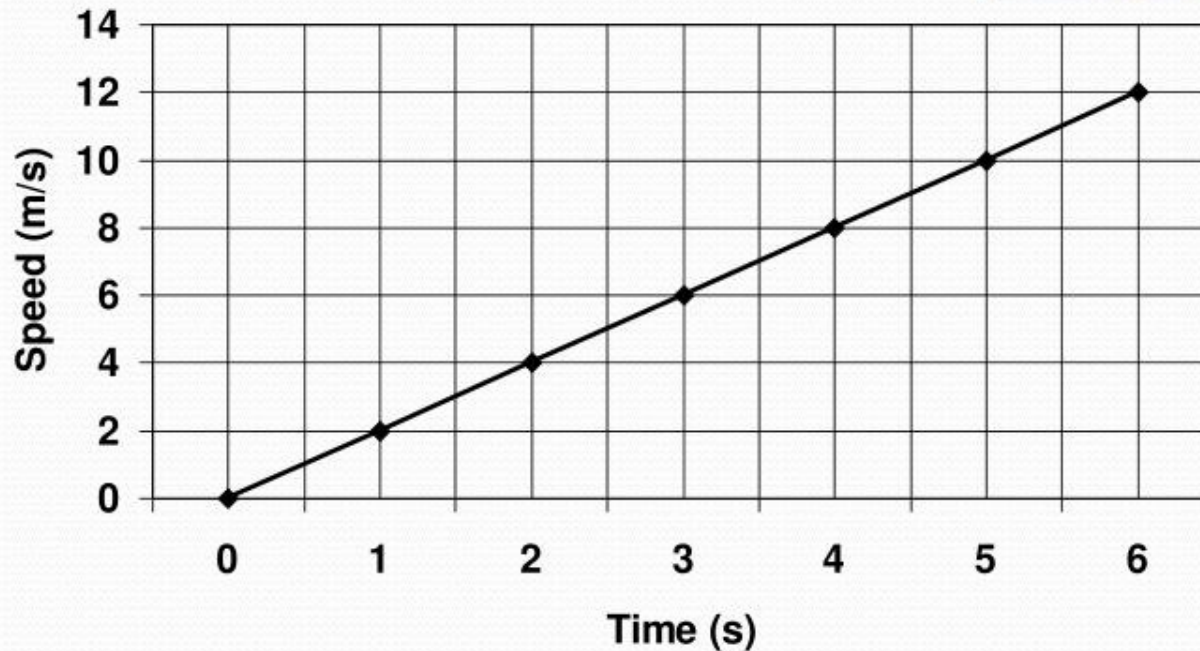
Question: Hard one



Above is a graph showing the speed of a car over time.

1) What would a distance vs. time graph for this look like?

Graphing Acceleration: Speed vs. Time Graphs



- 1) Speed is increasing with time = accelerating
- 2) Line is straight = acceleration is constant